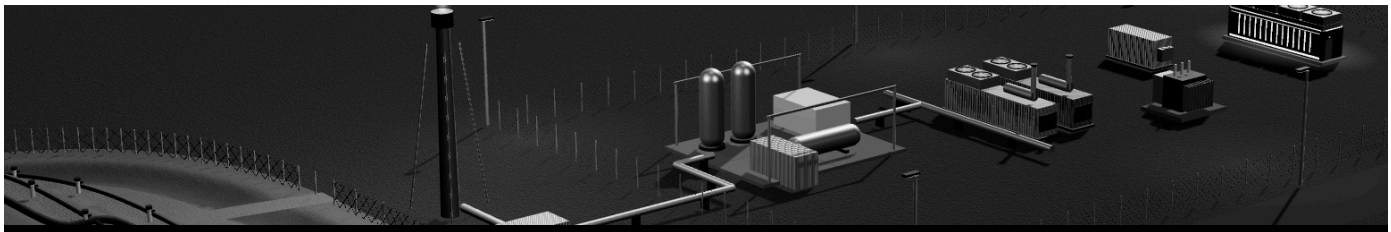
City of Charleston Landfill, Charleston, West Virginia: site sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

1.1 MW	1	6 ac	98k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $294k	$6k to $13k	$6k to $307k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=388&CH4=50.3

THE MATH

Landfill gas flared beside the RNG project (2023, estimate)	388 scfm
Methane content (2023)	50.3%
Energy in the gas	11.8 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 1.1 MW
Compute (power less cooling and losses, /1.2)	about 0.9 MW
Compute modules (about 1 MW each)	1
Gas engines (running plus one spare)	2 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 6 acres
Gas bought per year (95% run time)	98k MMBtu
Gas payment, up to $3.00/MMBtu	up to $294k
Land lease at $1,000 to $2,000 per acre per year	$6k to $13k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Our own gas engines on a fenced parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Measured flared flow today (this figure is an estimate)
- Gas beyond what the RNG project needs
- Space for a fenced parcel near the flare
- Gas quality (siloxanes, H2S) for engines

Source: Estimate from EPA data: about 858 scfm collected in 2023 at 50.3% methane (EPA Greenhouse Gas Reporting Program, subpart HH, facility 1002359: https://data.epa.gov/efservice/hh_gas_collection_system_detls/FACILITY_ID/1002359/JSON, seen 2026-11-13), less about 470 scfm to the RNG project (EPA LMOP database, September 2024 release: https://www.epa.gov/system/files/documents/2024-09/lmopcompositedata.xlsx). LMOP listed about 437 scfm flared in 2021. This is an estimate from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.